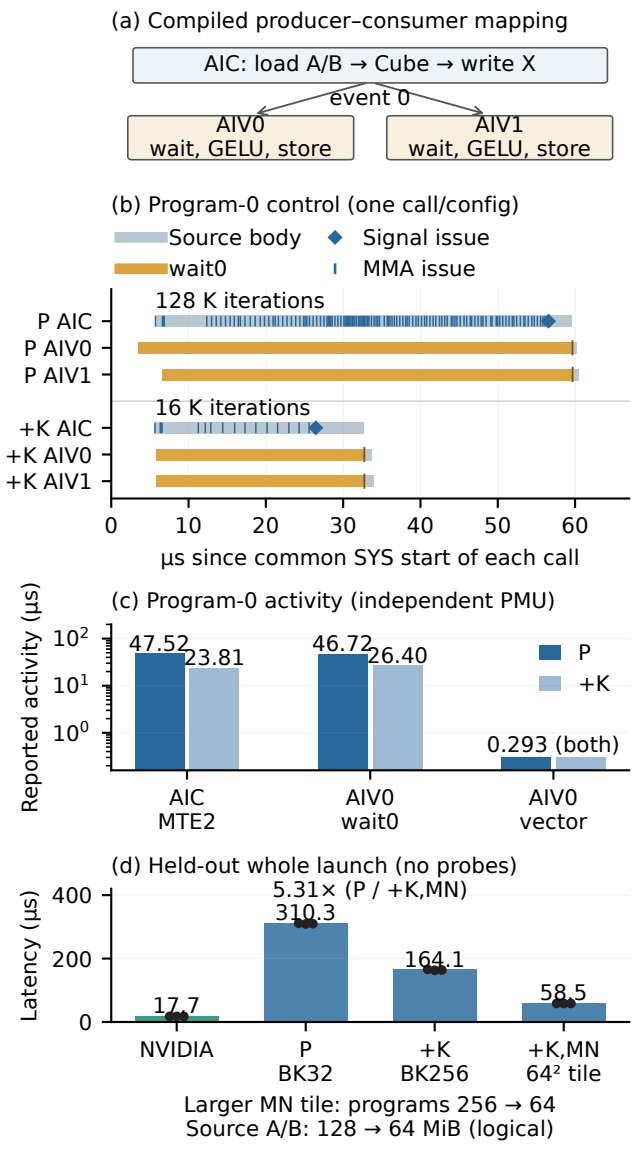
(a) Compiled producer–consumer mapping
AIC: load A/B → Cube → write X
event 0
AIV0
wait, GELU, store
AIV1
wait, GELU, store
(b) Program-0 control (one call/config)
Source body
Signal issue
wait0
MMA issue
128 K iterations
P AIC
P AIV0
P AIV1
16 K iterations
+K AIC
+K AIV0
+K AIV1
0
10
20
30
40
50
60
µs since common SYS start of each call
(c) Program-0 activity (independent PMU)
Reported activity (µs)
10^2
10^1
10^0
47.52
23.81
46.72
26.40
0.293 (both)
P
+K
AIC
MTE2
AIV0
wait0
AIV0
vector
(d) Held-out whole launch (no probes)
Latency (µs)
400
200
0
5.31× (P / +K,MN)
310.3
164.1
58.5
17.7
NVIDIA
P
BK32
+K
BK256
+K,MN
64² tile
Larger MN tile: programs 256 → 64
Source A/B: 128 → 64 MiB (logical)